Introduction to C Programming

What is C?

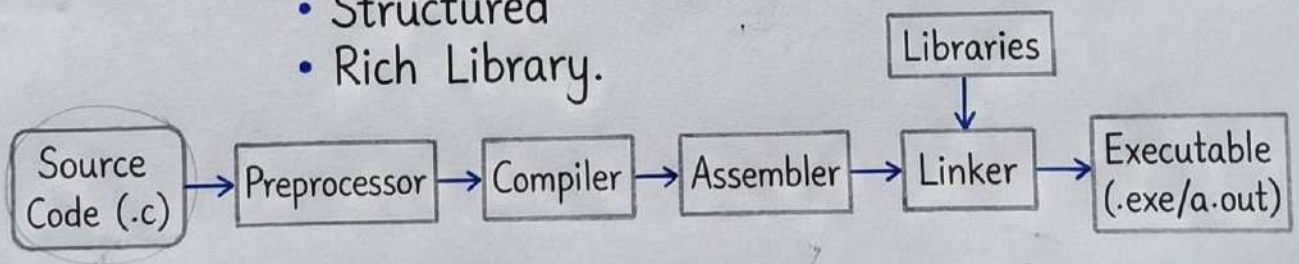
C is a general-purpose, procedural computer programming language supporting structured programming, lexical variable scope, and recursion, with a static type system. It was developed in the early 1970s by Dennis Ritchie at Bell Labs.

It is widely used for system programming, including developing operating systems (like UNIX), compilers, and embedded systems.

- Features: Simple
  - Efficient
  - Portable
  - Mid-level
  - Structured
  - Rich Library.

Exam Tip

Remember C is a foundational language for OS development & competitive coding!



Introduction to C Programming

History of C

- C was developed at Bell Labs in 1972.
- It was designed by Dennis Ritchie as an evolution of the B language.
- Derived from an earlier language called BCPL (Basic Combined Programming Language).
- Unix operating system was rewritten in C, demonstrating its power and efficiency.

Key Features of C

- **Procedural Language** – C follows the procedural paradigm, using functions to structure code.
- **Static Type System** – Every variable has a specific data type that is known at compile time.
- **Low-Level Access** – Allows manipulation of bits, bytes, and addresses for system-level programming.
- **Fast & Efficient** – C code is fast due to minimal runtime overhead, making it suitable for performance-critical applications.
- **Powerful Standard Library** – C has a rich set of built-in functions for performing a variety of common tasks.
- **Portability** – C programs can be compiled and run on many different platforms with little or no modification.

Introduction to C Programming

Basic Syntax of C

A C program consists of various components structured in a specific way. A simple C program has the following elements:

- **#include Statements** – Include standard library header files. (e.g. #include<stdio.h>
- **int main() Function** – The starting point of the program.
- **Braces { }** – Define the beginning and end of a function/block of code.
- **Statements** – Individual instructions end with a semicolon (:).
- **Comments** – Used for adding notes, ignored by the compiler. (e.g. // This is a comment).
- **Return Statement** – Ends the main() function and returns a value to the caller.
- **Semicolons** – Every statement ends with a semicolon (:).

```
#include<stdio.h>                    → #1 #include Statement
// Include standard library          → #2 main() Function
int main() {
    // Main function starts          → #3 Comment
    printf("Hello, World!\n");       → #4 Statement
    return 0; // Return 0 to the caller
    }                                → #5 Return Statement
}
```

Introduction to C Programming

Variables and Data Types

Variables in C must be declared before they can be used. A variable declaration specifies a data type and a name for the variable. Here are some common data types in C:

Data Type	Description	Size	Example
int	Integer type	4 bytes	int age = 25;
float	Floating-point number	4 bytes	float weight = 65.5f;
char	Single character	1 byte	char grade = "A";
double	Double precision floating-point.	8 bytes	double pi = 3.14159;
void	No value (used with functions)	–	–
_Bool	Boolean (true/false) value.	1 byte	_Bool isHappy = 1;

Note

C is case-sensitive. This means "Age", "AGE", and "age" would be considered different variables.

Variable Declaration

Syntax: `data_type variable_name = value;`

Example: `int age = 25;`

Helpful Tip

Use meaningful variable names. Eg, use "height" instead of "h" for better readability.

Introduction to C Programming

Input and Output in C

In C, input and output operations are carried out using standard library functions. Here are the primary functions used:

printf() Function

Usage: `printf(format_string, arguments);`

- Format specifiers begin with % to represent variable types within the format string.

Format Specifier	Integer	Example
%d	Floating-point	printf("D:\", %s\n", name);
%f	String	printf('%d", &age);

↳ Example:
```
int age = 25; float height = 5.9;
printf("Age: %d. Height: %.1f feet.\n", age, height);
```
Output: Age: 25, Height: 5.9 feet.

scanf() Function

Usage: `scanf(format_string, &variables);`

> **Exam Tip**
> Remember to use &" before variable names in "scanf" to provide the address of the variable for storing the input value.

```
#include<stdio.h>
// Include standard library
int main() {
    int age; // Declare an integer variable
    printf("Enter your age: ");// Output prompt
    scanf("%d", &age);
    printf("You are %d years old.\n", age; // Output result
    return 0;
}
```
Output Enter your age: 25
You are 25 years old.

Introduction to C Programming

Operators in C

C provides a variety of operators for performing operations on variables and values. Operators in C are categorized as follows:

Arithmetic Operators

Symbol	Description	Example
+	Addition	a + b
−	Subtraction	a − b
*	Multiplication	a * b
/	Division	a / b
%	Modulus	a % b

Relational Operators

Symbol	Description	Example
==	Equal to	a == b
!=	Not equal to	a != b
>	Greater than	a > b
<	Less than	a < b
>=	Greater than or equal to	a >= b

Logical Operators

Symbol	Description	Example
&&	AND	(a>0)&&(0>0)
||	OR	(a>0)||(b<0)
!	NOT	! (n == 0)

Assignment Operators

Symbol	Description	Example
=	Assign	a = 5
+=	Add and assign	a == 3 (a-a+3
−=	Subtract assign	a == 2 (a-a-2
+=	Multiply and asign	a == 4 (a-a*4
/=	Divide and assign	a/==2 (a-a/2

Example:

```
int a = 10, b = 5;
int sum = a + b;
if ((a > 0) && (b < 10)) {
    sum += 5;
    printf("Sum: %dIn", sum);  // Outputs: Sum: 20
}
Outputs: Sum: 20
```

Introduction to C Programming

Control Structures in C

Control structures allow you to manage the flow of a program. They include conditional statements and loops to control the execution of code blocks:

Conditional Statements

Conditional statements are used for decision-making.

if Statement	
Usage: `if (condition) { /* code */`	`if (x >0) { printf("Positive\n"); }`
if-else Statement	
Usage: `if (condition) { /* code if true */ /* code if false */`	`if (x >0) { printf("Positive\n"); }`
if-else if-else Statement	
Usage: `if (condition1) { /* code if condition1 is true */ /* code if condition2 is true */ /* code if all conditions are false *`	`if (x >0) { printf("Negative\n"); } else { printf("Zero\n"); }`

Loops:

Loops allow repetitive execution of a block of code. The primary loops in C.

• for Loop	Used to execute a block of code a specific number of times.
• while Loop	Executes a block of code as long as a condition is true.
• do-while Loop	Executes a block of code once, then repeats as long as condition is true.

Note:

Control structures are fundamental for writing flexible and efficient programs.

Introduction to C Programming

Loops in C

Loops are used to execute a block of code repeatedly. In C, there are three primary types of loops:

- for Loop
- for Loop
- while Loop

for Loop

Usage: for (initialization; condition; increment) {

- Example

Output: 0 1 2 3 4

```
for (int i = 0; i < 5; i++)
  printf("%d , i);
}
```

Output: 0 1 2 3 4

while Loop

Usage: while (condition) {

- Example:

Output: 0 1 2 3 4

```
int i = 0;
while (i < 5) {
  printf("%d , i);
  i++;
}
```

do-while Loop

Loops allow repetitive loop, the do-while loop guarantees at least one iteration of the code block.

- Example:

Output: 0 1 2 3 4

```
int i = 0;
do {
  printf("%d , i);
  i++;
} while (i < 5);
```

Output: 0 1 2 3 4

Tip

Uniike the while loop, the do-while loop guarantees at least one iteration of the code block.

Introduction to C Programming

Functions in C

Functions are used to break down programs into smaller, reusable pieces of code. They help to organize and manage code efficiently:

Syntax:

```
return_type function_name(parameter1, parameter2, ...) {
                                                    /* code */
    return value;
}
```

A function in C has the following components:

• Return Type:	Specifies the type of value the function returns (eg., int, void)
• Function Name:	The identifier used to call the function (e.g, sum)
• Parameters:	Inputs to the function, specified inside parentheses (optional)
• Return Value:	The value the function sends back to the caller (optional)

Example:

Usage: while (condition) {

• Example:

Output: 0 1 2 3 4

```
int i = 0;
while (i < 5) {
  printf("%d , i);
 i++,
}while ( i < 5);
```

do-while Loop

Usage: do { /* code */
} while (condition);

Output: 0 1 2 3 4

```
int i = 0;
 printf("%d , i);
i+++;
}while ( i < 5);
```

Tip

Uniike the while loop, the do-while loop guarontees at least one iteration of the code block.

Introduction to C Programming

Arrays in C

Arrays are used to store multiple values of the same data type. An array is a collection of elements that can be accessed using an index.

Syntax:

```
data_type array_name[size];                    /* code */
```

A function in C has the following components:

• Return Type:	Specifies the type of value the function returns (eg, int, float)
• array_name:	The identifier used to call the array
• size:	Inputs to the function, specified inside parertheses (optional)

Example:

Syntax: `data_type array_name[size] = {value1, value2, ...};`

```
int numbers[5] = {10, 20, 30, 40, 50};
```

Output: 0 1 2 3 4

10	20	30	40	50	50
0	1	2	3	3	4

Example:

Usage: `data_type array_name[size] = {value1, value2, ...};`

```
int numbers[5] = {10, 20, 30, 40, 50};
printf('Second element: %d \n', numbers[1]); // Access the second element.
```

Output: 0 1 2 3 4

Note:

Array indexing starts at 0, so the first element is at index 0, the second at index 1, and so on.

Introduction to C Programming

More on Arrays

Arrays in C can be maniplated in various ways. Here are some common operations and concepts related to arrays:

Syntax:

Array Initialization:	Arrays can be intialized at the time of declaration.
◦ Syntax:	`data_type array_name[size] = { value1, value2, ... };`

- A function in C has the following components:

◦ data_type:	The data type of the elements (e.g., int, float)
◦ array_name:	The identifier used to call the array

Example:

Syntax: `data_type array_name[size] = { value1, value2, ... };`

```
int numbers[5] = {10, 20, 30, 40, 50};
                                    // Garbage: Int *V value - 87; index = 0
```

Output: 0 1 2 3 4

Array Length

The length (number of elements) of an array can be determined using:

```
length = sizef(array_name) / sizef(data_type);
```

```
int numbers[5] = {10, 20, 30, 40, 50);
int length = sizef(numbers) / sizef( numbers[0]);
printf("Array length: %d\n", length);
```

Output: Array length: 5

Note:

Be careful when accessing arrays. Accessing an index out of bounds (e.g., numbers[5]) can lead to undefined behavior or crashes.

Introduction to C Programming

Pointers in C

Pointers are variables that store the memory address of another variable. They are used for dynamic memory allocation, arrays, and functions.

Syntax:

```
data_type *pointer_name;                    /* code */
```

- A function in pointer selection various ways:

• Syntax:	The data type of the variable the pointer (e.g., int, float)
* *:	Indicates the indicates variable) a pointer!
• pointer_name:	The identifier used to call the pointer

Example:

```
int num = 10 ;                    ] = Pointer for is & num
int *ptr = &num; // Pointer to num
.printf((Value of num: %d\n., num);
.printf((Value pointed to by ptr: %d\n, *ptr);
```

Output: Value of num: 10

Value pointed to by ptr: 10

Pointer and Address-of Operator

The address-of operator (&) gives the memory address of a variable.

```
Dointer:
Tht numbers[5] = {10, 20, 30, 40};
prt memors numbers[1]; /aimuee, ahd.
```

Output: Array length: 5

Deefereencing Pointer

The dereferencing operator (*) accesses the value at the pointers address.

```
prt = *tener(array_namer)-
    aiemer{ stie_sprte;
printf('Array length, %d\n, length);
```

Output: Array length: 5

Introduction to C Programming

Dynamic Memory Allocation

Dynamic memory allocation allows programs to request and manage memory while the program is running. In C, we use the stdlib.h library functions to allocate and free memory dynamically:

Syntax:

malloc() Allocates a specfied number of bytes and returns a pointer to the allocated memory.

- Syntax: `pointer = (data_type*) malloc(size_in_bytes);`

- A function in C has the following components:

malloc()
p.

```
int * arr ;
arr = (int*) malloc(5 * sizof(int)); // Dynamically allocate an array
if (arr != NULL);
    arr[0] = 10;
    printf("First element : %d\n", arr[0]);
free(arr); // Free dynamically allocated memory
```

Output: First element: 10

Pointer and Address-of Operator

The address-of operator (&) gives the memory address of a variable.

Syntax:

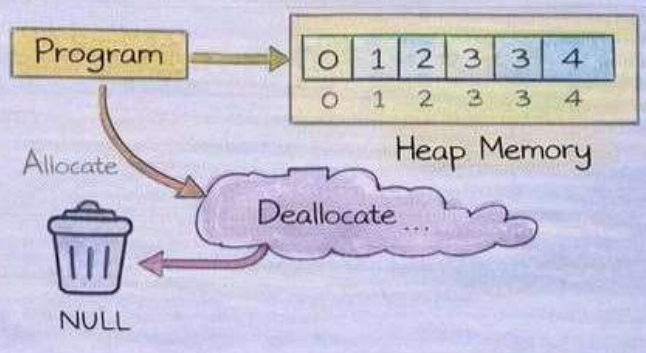
```
free(pointer_name);
```

Output: First element: 10

Note:
Failing to free dynamically allocated memory can lead to memory leaks.

Dereferencing Pointer

Frees dynamically allocated memory, releasing it for future use.



Introduction to C Programming

More on Dynamic Memory Allocation

Beyond malloc() and free(), the realloc() function can be used to resize previously allocated memory blocks.

Syntax:

```
pointer = realloc(pointer, new_size);                    / f code */
```

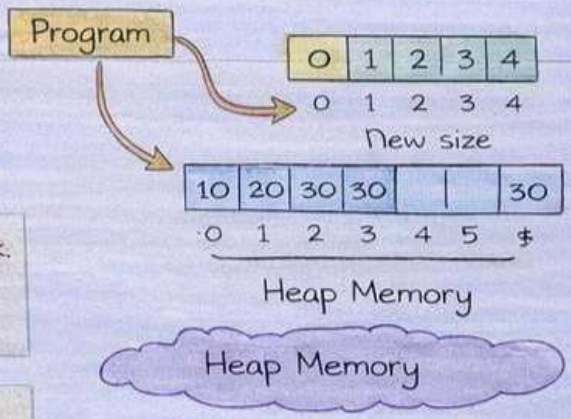
- A function in C has the following components:

- Syntax: `pointer = (data_type*) malloc(size_in_bytes);`

- A function in C has the following components:

```
int * arr;
arr = (int*) malloc(3 * sizof(int)); // Allocate an array of 3 integers
if (arr != NULL);
    arr[0] = 10;
    arr[2] = 20;
    arr = realloc.(arr, 5 * sizózof(int)). // Resize the array to hold 5 integers
    if (arr != NULL);{
        arr[3] = 40;
        arr[4] = 50;
    printf("Elements in the array: ");
Output: First element: 10
```

Pointer "arr" points here



Tip: Always check if realloc() successfully reallocated the memory before using it.

Tip: realloc() can be used to grow or shrink an array of dynamically allocated memory to a new size.

Note:
Foiling to free dynamically allocated memory can lead to memory leaks.

Introduction to C Programming

More on Dynamic Memory Allocation

Beyond malloc() and free(), the realloc() function can be used to resize previously allocated memory blocks:

if Statement:

- Executes a block of code if a condition is true.

Syntax: `pointer = realloc(pointer, new_size);` /* code */

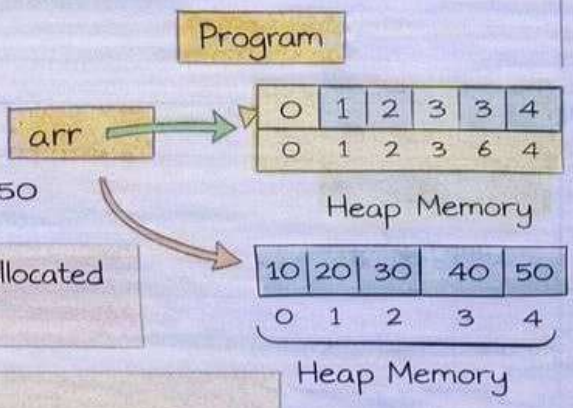
- A function in C has the following comporents:

- Syntax: `pointer = realloc( pointer, new_size);`

Example:

```
int num = 10;
if (num > 0){    // Allocate an arrag of 5 integers
    arr[0] = 10;
    arr[2] = 20;
    arr = realloc (arr, 5* sizof(int)); // Resize the array to hold 5 integers
    if (arr != NULL);
        arr[3] = 40;   <-- Pointer "arr" points here
        arr[4] = 50;
     printf("Elements in the array: ");
for (int i =.0 , < 5 );
free (arr);
```

Output: Elements in the array: 10 20 30.40 50



Tip: Always check if realloc() successfully reallocated the memory before using it.

Note:

It's important to use proper conditionals to control the flow of a program and handle different situations.

Introduction to C Programming

Examples of Conditionals

Here are three examples illustrating the use of conditional statements in C.

if Statement

- Executes a block of code if a condition is true.

Syntax: `pointer = realloc(pointer, new_size);`

```
int num = 15;
if (num > 0){
    printf("The number is positive.\n");
}
```

Output: The number is positive.

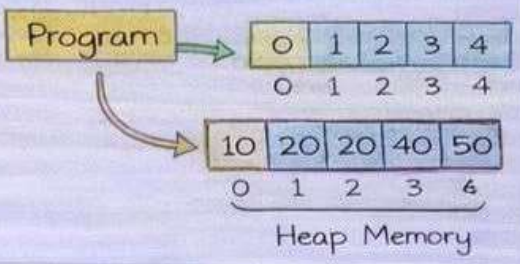
if-else Statement

- Executes one block of code if a condition is true, and another block if it is false.

Syntax:

```
int num = 42;
if (num % 2 == 0){
    printf("The number is even.\n");
  } else {
    printf("The number is odd.\n");
}
```

Output: Non positive.



if-else if-else Statement:

- Example: Check whether a number is positive, negative, or zero.

Tip:

Consider all possible conditions to ensure your program behaves correctly under different inputs.

Introduction to C Programming

The switch Statement

Here are three examples illustrating the use of conditional statements in C.

if Statement

- Executes a block of code if a condition is true.

Syntax: switch (variable) / f code */

- A faction name can to print the day number.
- Switch (variable) {

```
case value1:
    (f code for value1 }
    break;
}
```

Output: The number is positive.

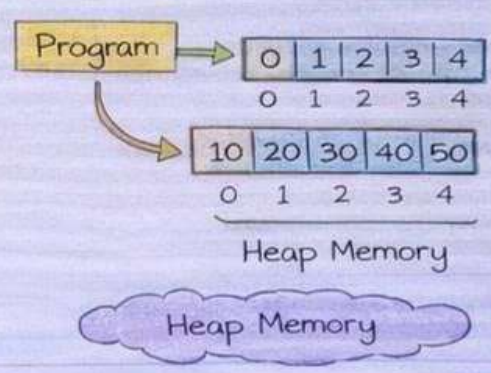
if-else Statement:

- Executes one block of code if a condition is true, and another block if it is false.

Syntax:

```
int num = 42;
if (num % 2 == 0){
    printf("The number is even.\n");
} else {
    printf("Non positive \n");
}
```

Output: Wednesday



Output: Wednesday

Note:

Its impuder all possible conditions to ensure your program behaves correctly under different inputs.